

Project Design Phase

Solution Architecture

Date	29 September 2026
Team ID	SWTID-2026-9139
Project Name	AI WeatherWise
Maximum Marks	4 Marks

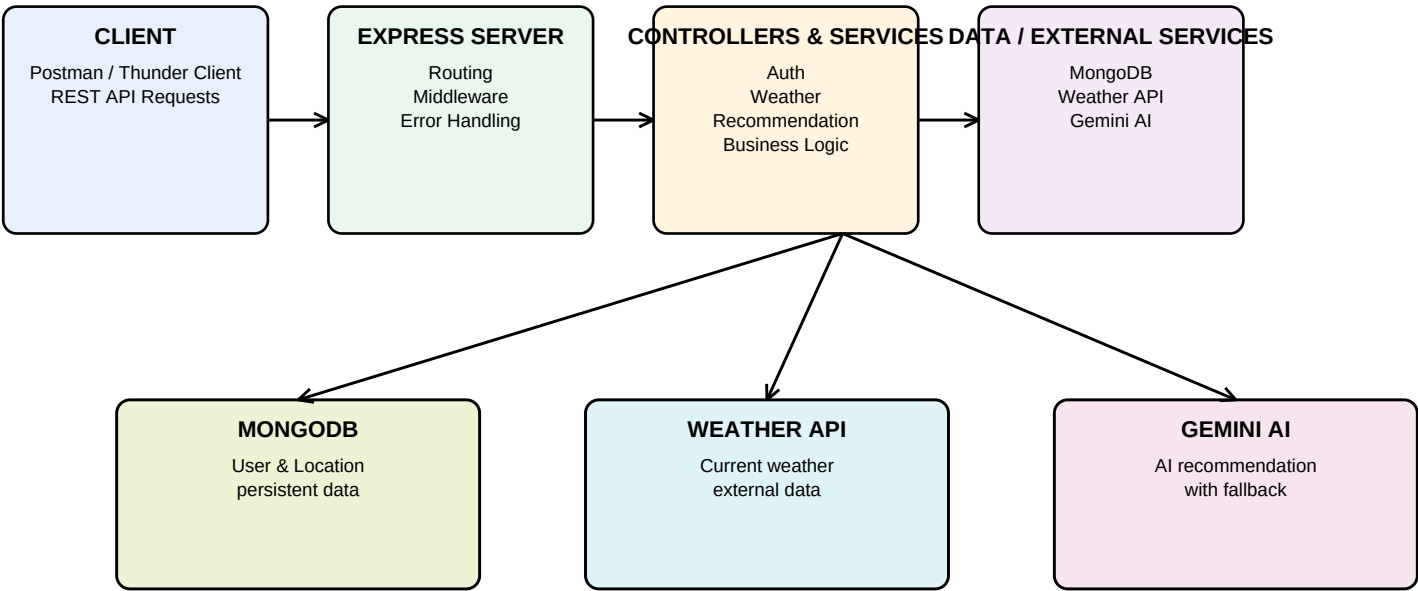
Solution Architecture:

Solution architecture bridges the project problem and its technology implementation. For AI WeatherWise, the architecture organizes API requests through an Express server, authentication and request-processing middleware, controllers and services, persistent MongoDB data, and external Weather API and Gemini AI services.

Goals:

- Define the structure and behaviour of the backend solution.
- Show how client requests move through the backend layers.
- Identify database and external-service integrations.
- Provide a clear technical structure for development and maintenance.

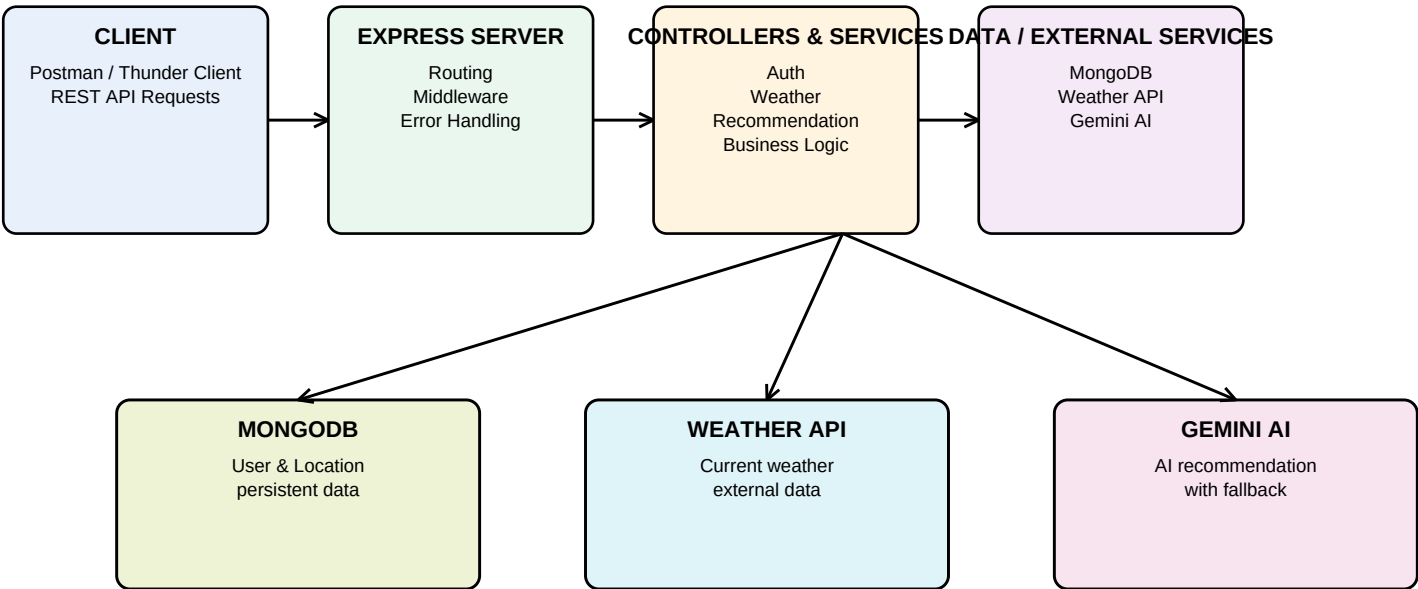
Example - Solution Architecture Diagram:



AI WeatherWise — Solution Architecture

System Architecture and Data Flow

The following architecture is the project-specific representation for AI WeatherWise. It follows the structure used in the project materials: Client → Express Server → Middleware / Controllers → Services → MongoDB / Weather API / Gemini AI.



Architecture Components

Client	API client such as Postman or Thunder Client sends REST requests.
Express Server	Node.js/Express handles routing, middleware, request processing and centralized error handling.
Middleware & Controllers	Authentication, input processing and controller workflows handle application requests.
Services Layer	Weather and recommendation services communicate with external services and apply backend logic.
MongoDB	Mongoose models persist user profiles and favorite locations.
External Services	Weather API supplies current weather data; Gemini AI supports weather-based recommendations, with fallback support.

Architecture Flow: Client request → Express routing → middleware/controllers → service logic → MongoDB and/or external Weather API/Gemini AI → response returned to the client.